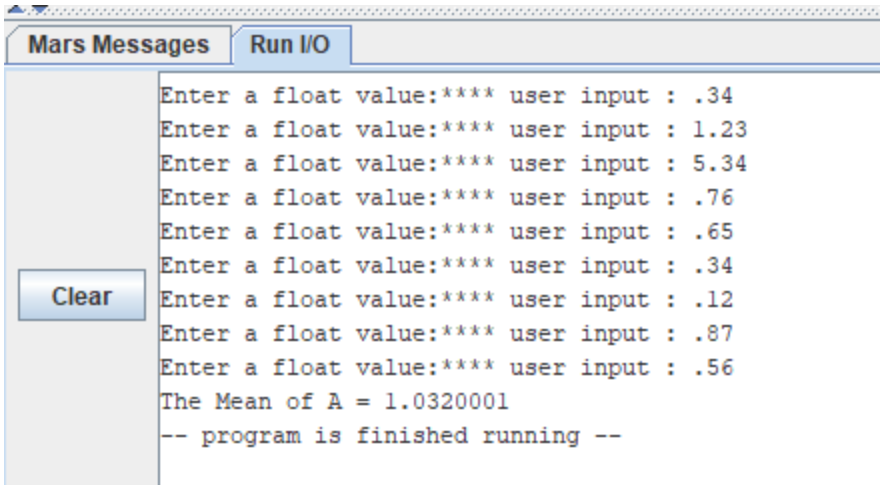
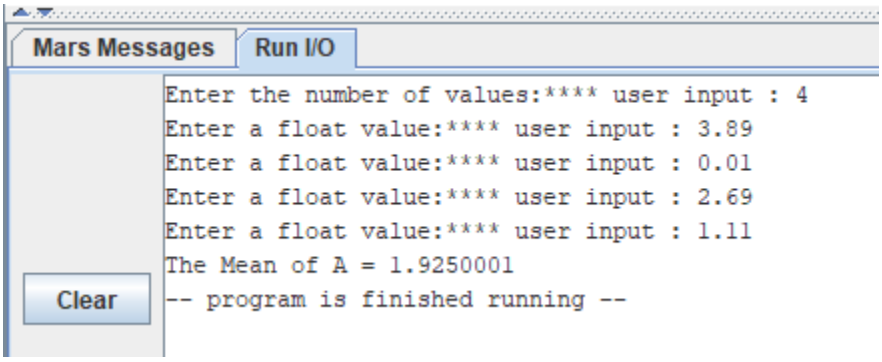
Project 3 CO Report

1. Austin Sternberg
2. Code File: Project3_average.asm
3. This program prompts a user to ask how many float numbers they are going to enter, then asks for those numbers. From there it calculated the mean or average of those floating numbers and outputs it. Lines 10-19 of this program ask for that integer, then store it in register $t0. Lines 20-23 move this value to register $f6 and convert it to a foat with the cvt.s.w command. The lwcl $f4, zeroFloat instruction is used to initialize the sum register to zero before reading in the float numbers. The loop, defined between lines 28-46, checks the value in $t0. If it is not zero, then the syscall section will prompt the user for a float number, read that value, and add it to the sum value in register $f4. The loop will then decrement the value in $t0 by one and jump back to the top. When the value of $t0 equals zero, the program will break out of the loop and go to line 48, where it will use div.s to divide the sum value by the converted integer in $f6, storing this in $f8. The final part of the program, lines 51-64 will print this average value out to console, and then gracefully exit.

4. 



5. This project was pretty straightforward. It was helpful to have the farh-to-cel.asm file as this made the division part far, far simpler to implement. However, figuring out how to convert the integer value to a float was difficult. I had to spend some time researching how to do so and came across the cvt.s.w instruction. The instruction is the basis of how this whole program can actually work, yet it is fairly simple to understand. At this point I am pretty confident with loops in MIPS, though I don't know if my implementation of reading everything in as an addendum to the current value of the sum is the best approach. It may make more sense to store all of the values individually, then add them together at the end. I suppose this would limit the maximum number of float values that could be entered though. In either case, this project taught me how to deal with different data types and how to convert them.